

Introduction to Fuzzy Logic Control

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Overview

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- Introduction
- Crisp Variables
- Fuzzy Sets
- Linguistic Variables
- Membership Functions
- Fuzzy Logic
 - Fuzzy OR
 - Fuzzy AND
 - Example
- Fuzzy Control
 - Variables
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 - Fuzzification
 - Defuzzification
- Summary

- Outline to the left in green
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- Case Study

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- L. Zadah, “Fuzzy sets as a basis of possibility” Fuzzy Sets Systems, Vol. 1, pp3-28, 1978.
- T. J. Ross, “Fuzzy Logic with Engineering Applications”, McGraw-Hill, 1995.
- K. M. Passino, S. Yurkovich, "Fuzzy Control" Addison Wesley, 1998.

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- Fuzzy logic:
 - A way to represent variation or imprecision in logic
 - A way to make use of natural language in logic
 - Approximate reasoning
- Humans say things like "If it is sunny and warm today, I will drive fast"
- Linguistic variables:
 - Temp: {freezing, cool, warm, hot}
 - Cloud Cover: {overcast, partly cloudy, sunny}
 - Speed: {slow, fast}

Crisp (Traditional) Variables

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- Crisp variables represent precise quantities:
 - $x = 3.1415296$
 - $A \in \{0,1\}$
- A proposition is either True or False
 - $A \wedge B \Rightarrow C$
- $\text{King}(\text{Richard}) \wedge \text{Greedy}(\text{Richard}) \Rightarrow \text{Evil}(\text{Richard})$
- Richard is either greedy or he isn't:
 - $\text{Greedy}(\text{Richard}) \in \{0,1\}$

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- What if Richard is only somewhat greedy?
- Fuzzy Sets can represent the degree to which a quality is possessed.
- Fuzzy Sets (Simple Fuzzy Variables) have values in the range of $[0,1]$
- $\text{Greedy}(\text{Richard}) = 0.7$
- Question: How evil is Richard?

Fuzzy Linguistic Variables

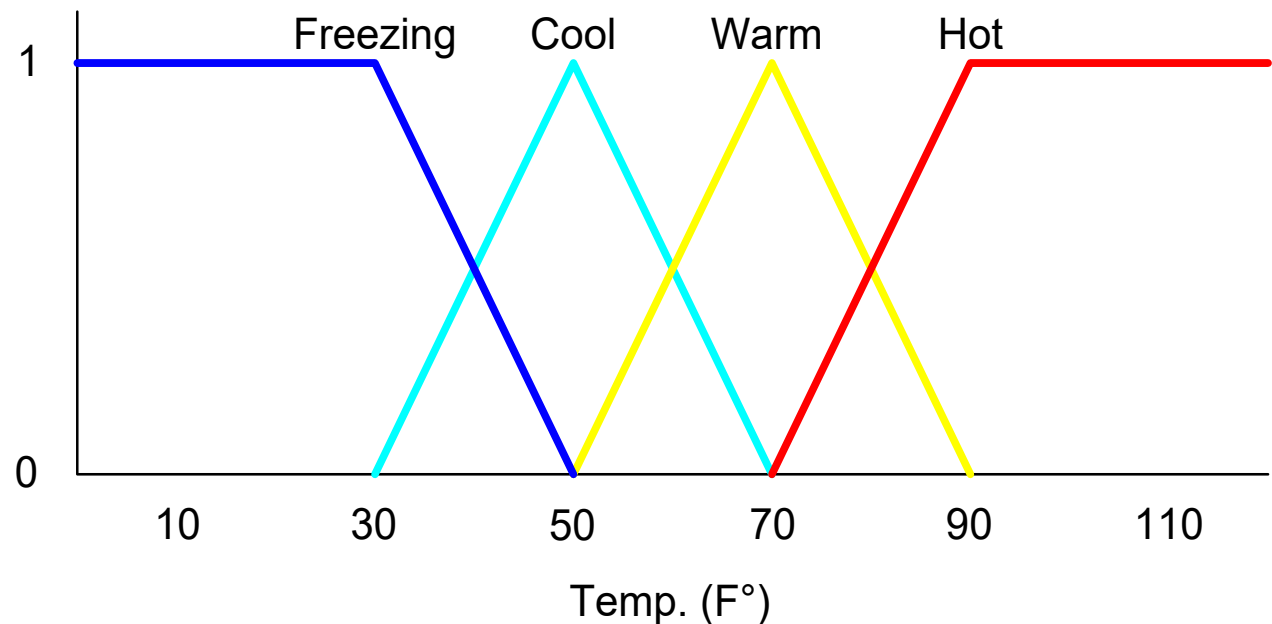
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- Fuzzy Linguistic Variables are used to represent qualities spanning a particular spectrum
- Temp: {Freezing, Cool, Warm, Hot}
- Membership Function
- Question: What is the temperature?
- Answer: It is warm.
- Question: How warm is it?

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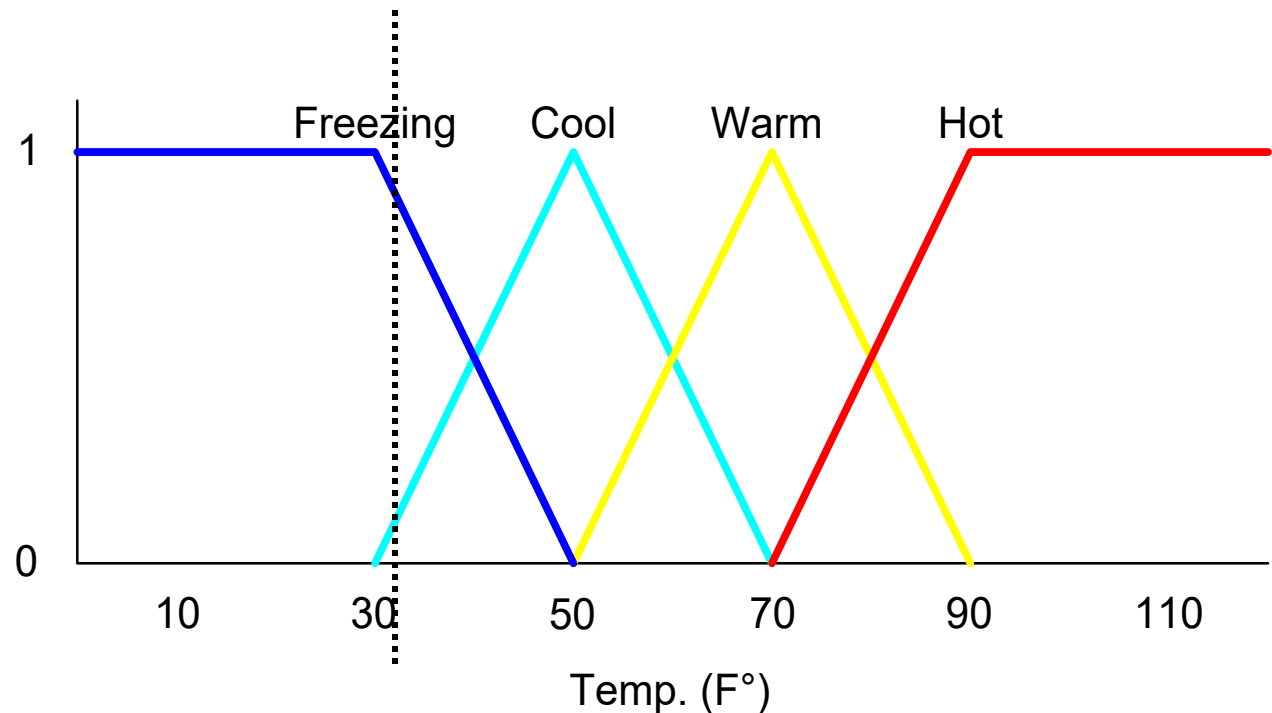
- Temp: {Freezing, Cool, Warm, Hot}
- Degree of Truth or "Membership"



Membership Functions

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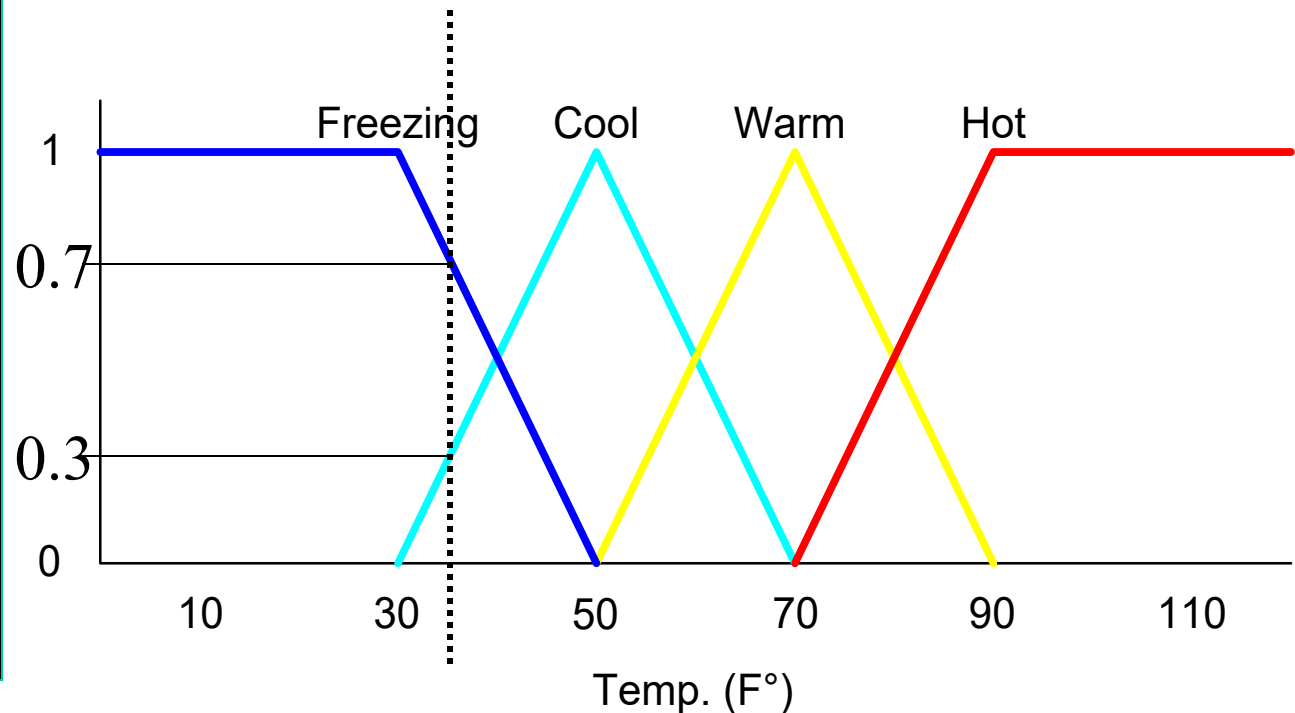
- How cool is 36 F° ?



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- How cool is 36 F° ?
- It is 30% Cool and 70% Freezing



Fuzzy Logic

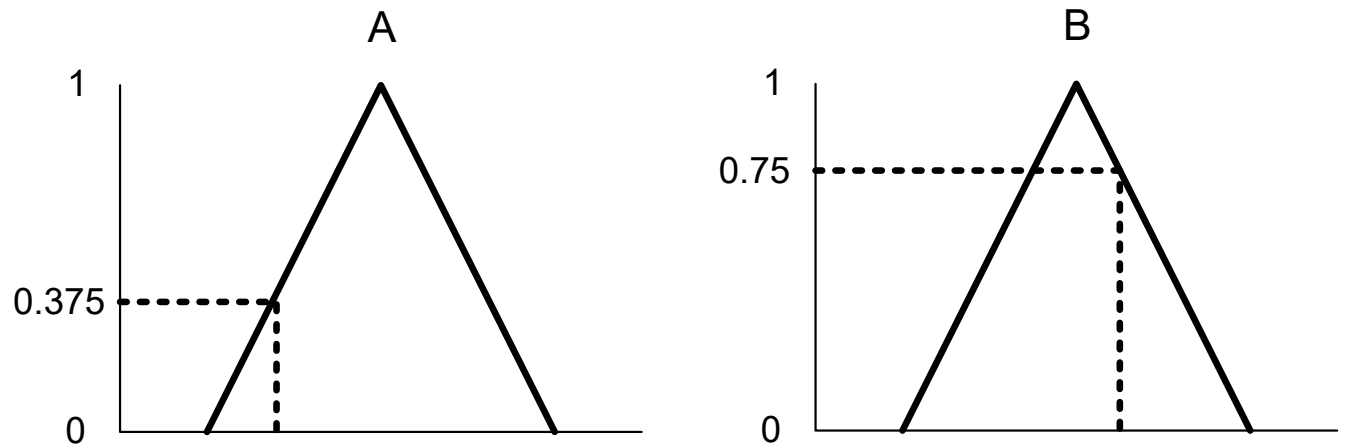
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- How do we use fuzzy membership functions in predicate logic?
- Fuzzy logic Connectives:
 - Fuzzy Conjunction, $\wedge$
 - Fuzzy Disjunction, $\vee$
- Operate on degrees of membership in fuzzy sets

Fuzzy Disjunction

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- $A \vee B \triangleq \max(A, B)$
- $A \vee B = C$ "Quality C is the disjunction of Quality A and B"

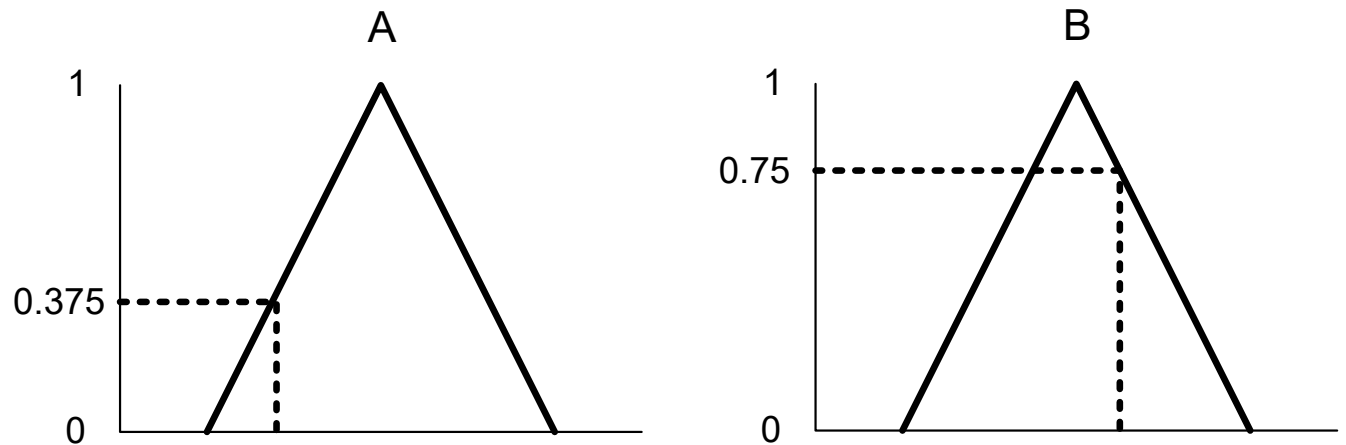


- $(A \vee B = C) \Rightarrow (C = 0.75)$

Fuzzy Conjunction

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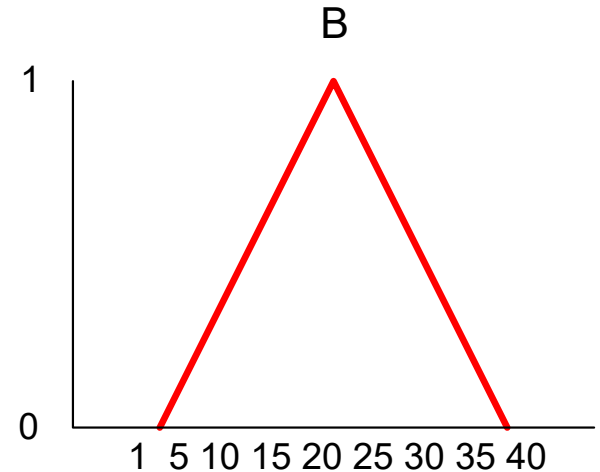
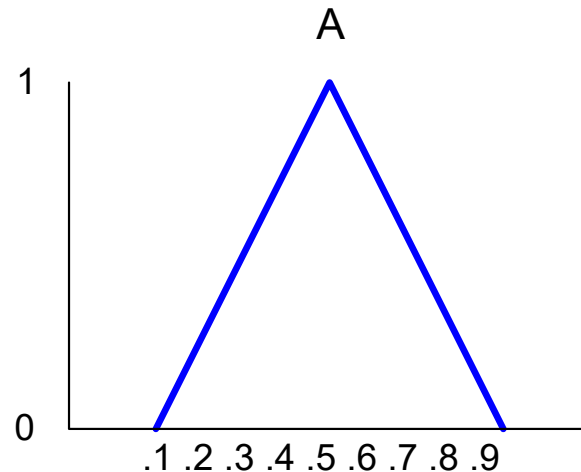
- $A \wedge B \triangleq \min(A, B)$
- $A \wedge B = C$ "Quality C is the conjunction of Quality A and B"



- $(A \wedge B = C) \Rightarrow (C = 0.375)$

Example: Fuzzy Conjunction

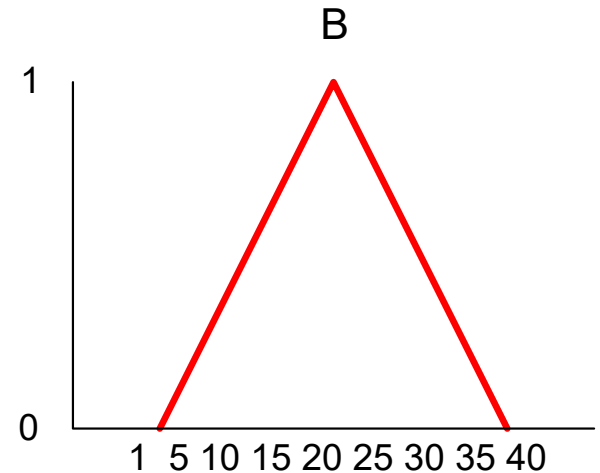
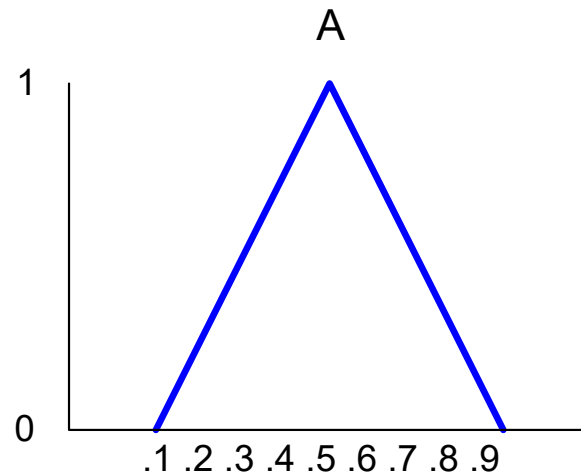
Calculate $A \wedge B$ given that A is .4 and B is 20



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Example: Fuzzy Conjunction

Calculate $A \wedge B$ given that A is .4 and B is 20

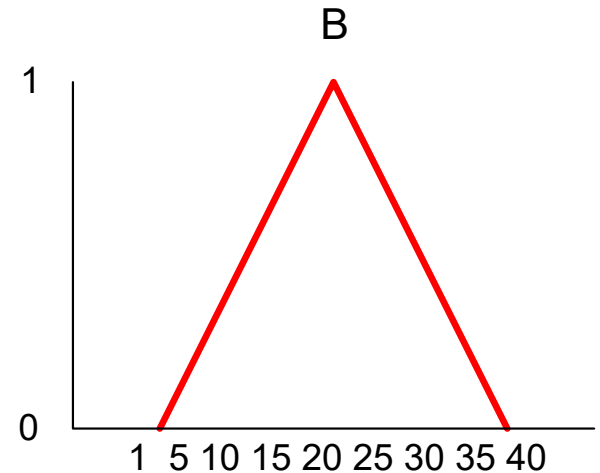
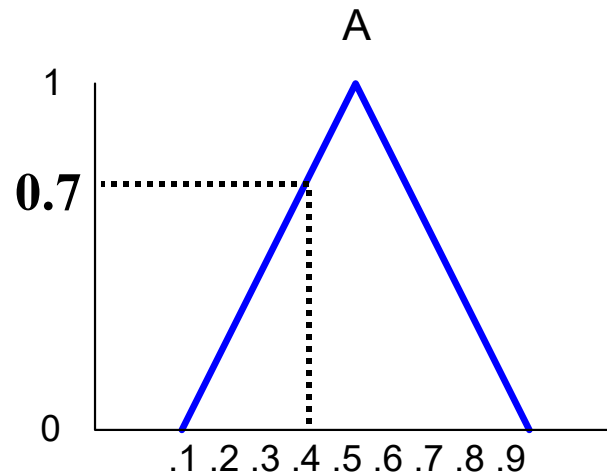


- Determine degrees of membership:

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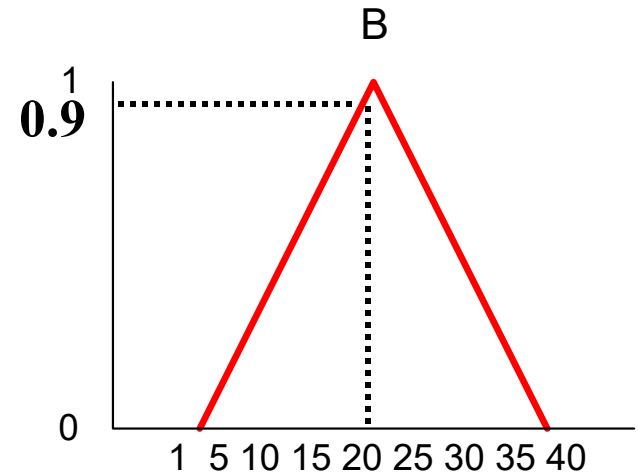
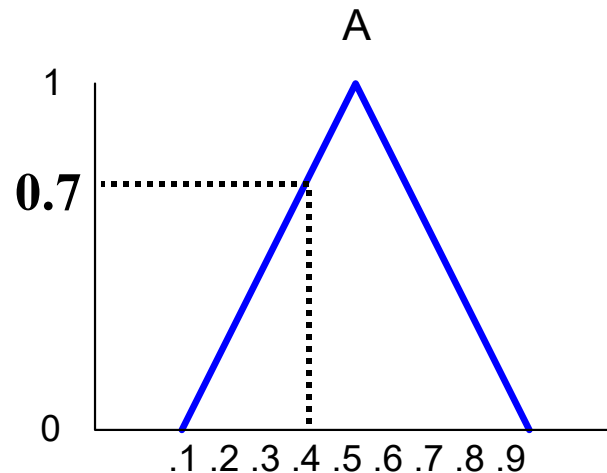


- Determine degrees of membership:
 - $A = 0.7$

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Example: Fuzzy Conjunction

Calculate $A \wedge B$ given that A is .4 and B is 20

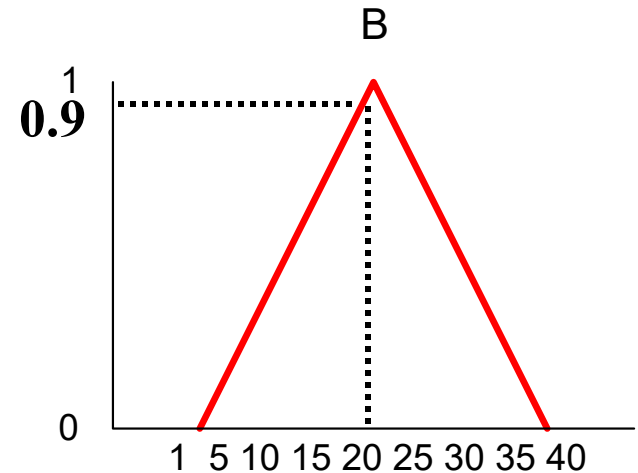
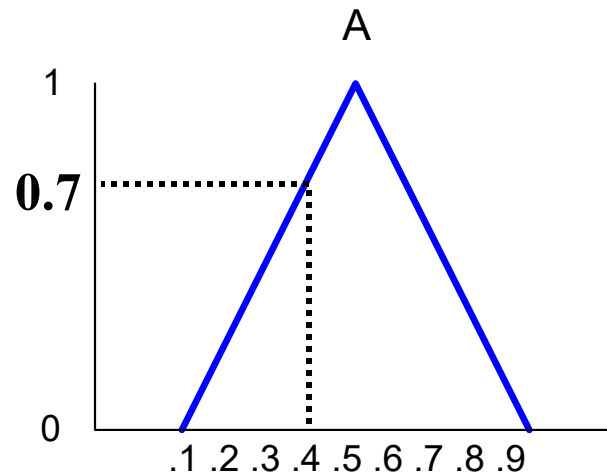


- Determine degrees of membership:
 - $A = 0.7$ $B = 0.9$

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Example: Fuzzy Conjunction

Calculate $A \wedge B$ given that A is .4 and B is 20



- Determine degrees of membership:
 - $A = 0.7$ $B = 0.9$
- Apply Fuzzy AND
 - $A \wedge B = \min(A, B) = 0.7$

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Fuzzy Control

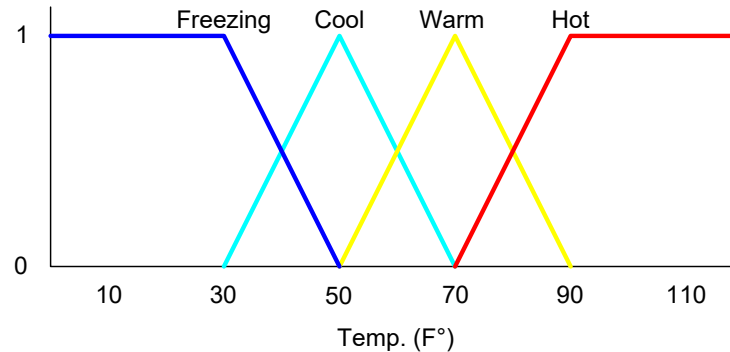
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- Fuzzy Control combines the use of fuzzy linguistic variables with fuzzy logic
- Example: Speed Control
- How fast am I going to drive today?
- It depends on the weather.
- Disjunction of Conjunctions

Inputs: Temperature

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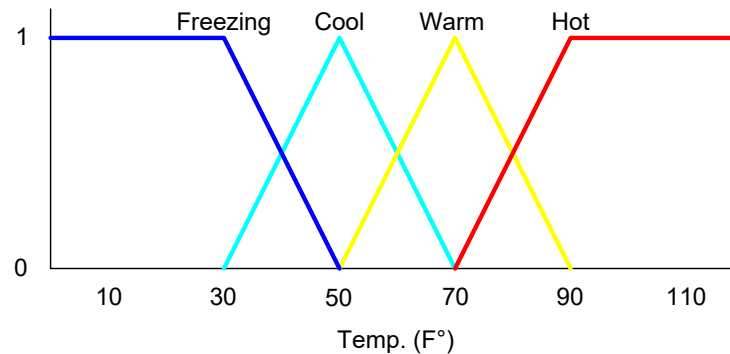
- Temp: {Freezing, Cool, Warm, Hot}



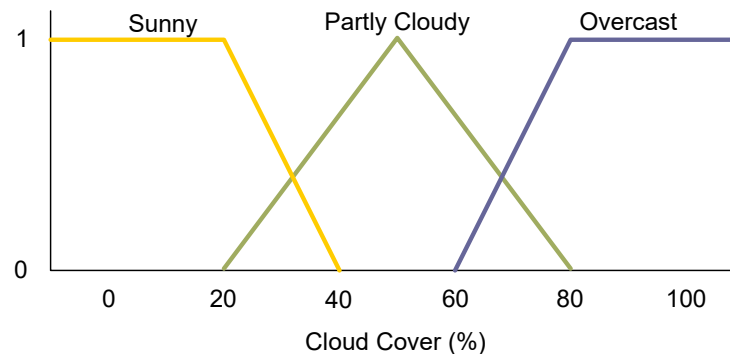
Inputs: Temperature, Cloud Cover

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- Temp: {Freezing, Cool, Warm, Hot}

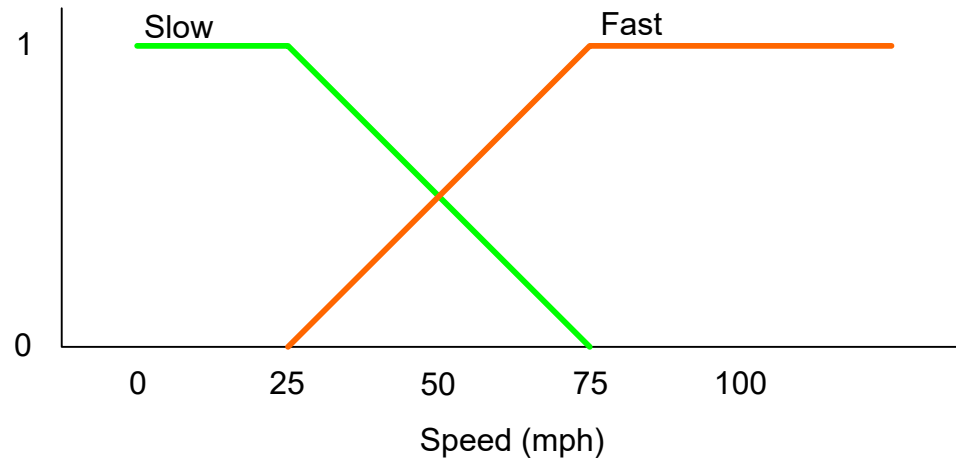


- Cover: {Sunny, Partly, Overcast}



Output: Speed

- Speed: {Slow, Fast}



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- If it's Sunny and Warm, drive Fast
 $\text{Sunny}(\text{Cover}) \wedge \text{Warm}(\text{Temp}) \Rightarrow \text{Fast}(\text{Speed})$
- If it's Cloudy and Cool, drive Slow
 $\text{Cloudy}(\text{Cover}) \wedge \text{Cool}(\text{Temp}) \Rightarrow \text{Slow}(\text{Speed})$
- Driving Speed is the combination of output of these rules...

Example Speed Calculation

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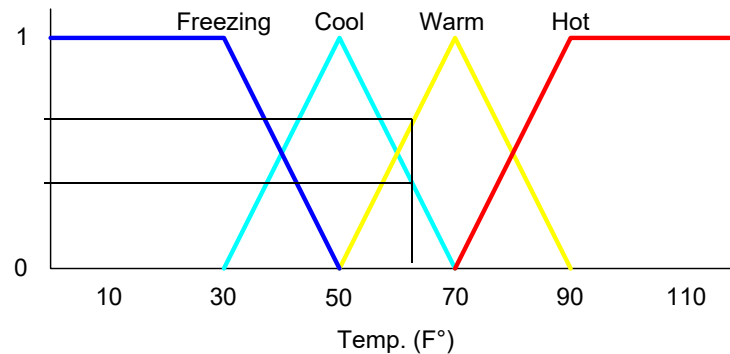
- How fast will I go if it is
 - 65 F°
 - 25 % Cloud Cover ?

Fuzzification:

Calculate Input Membership Levels

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- $65\text{ F}^\circ \Rightarrow \text{Cool} = 0.4, \text{Warm} = 0.7$

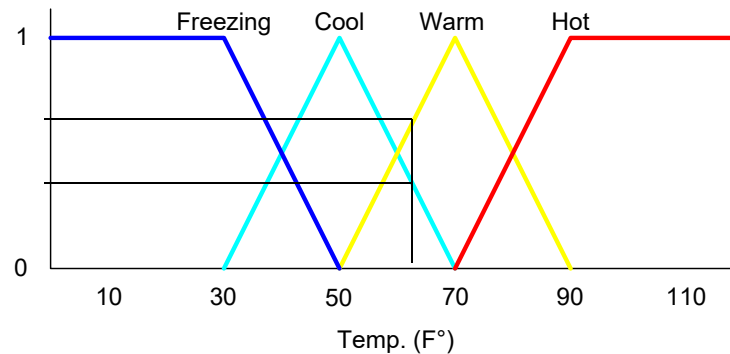


Fuzzification:

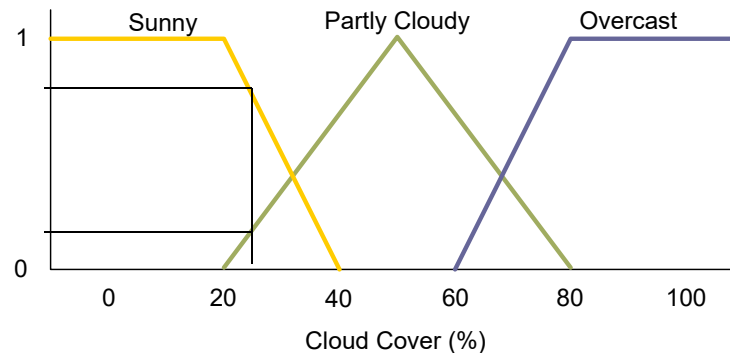
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- $65\text{ F}^\circ \Rightarrow \text{Cool} = 0.4, \text{Warm} = 0.7$



- $25\% \text{ Cover} \Rightarrow \text{Sunny} = 0.8, \text{Cloudy} = 0.2$



...Calculating...

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- If it's Sunny and Warm, drive Fast

$\text{Sunny}(\text{Cover}) \wedge \text{Warm}(\text{Temp}) \Rightarrow \text{Fast}(\text{Speed})$

$$0.8 \wedge 0.7 = 0.7$$

$$\Rightarrow \text{Fast} = 0.7$$

- If it's Cloudy and Cool, drive Slow

$\text{Cloudy}(\text{Cover}) \wedge \text{Cool}(\text{Temp}) \Rightarrow \text{Slow}(\text{Speed})$

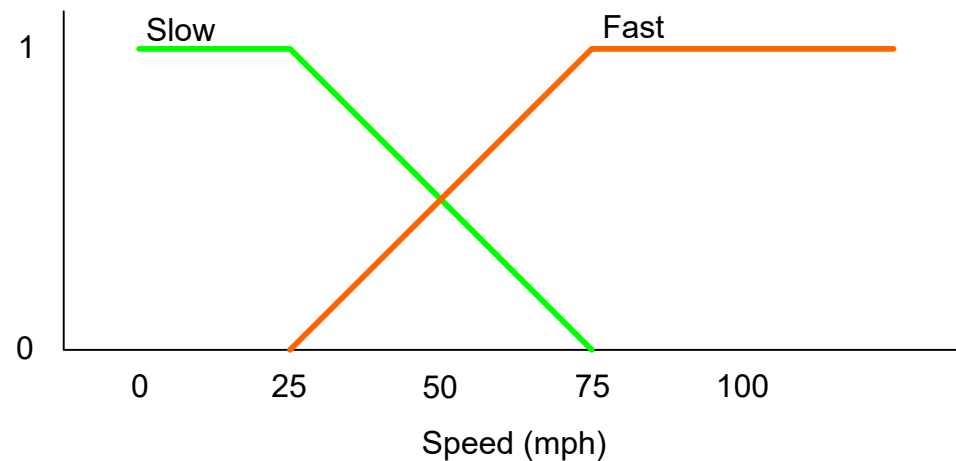
$$0.2 \wedge 0.4 = 0.2$$

$$\Rightarrow \text{Slow} = 0.2$$

Defuzzification: Constructing the Output

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- Speed is 20% Slow and 70% Fast

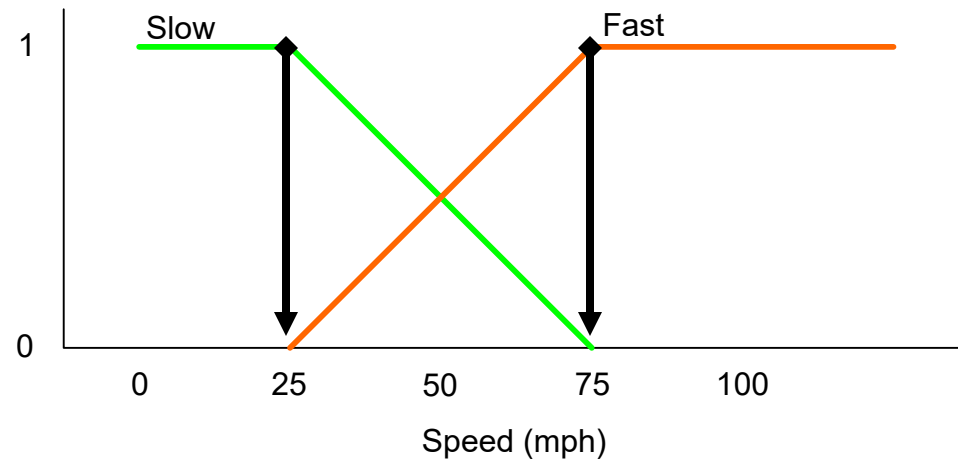


- Find centroids: Location where membership is 100%

Defuzzification: Constructing the Output

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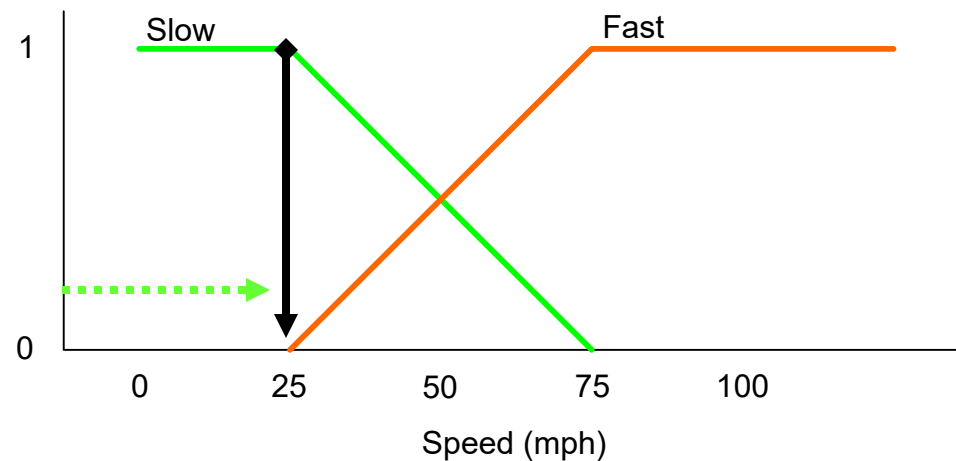


- Find centroids: Location where membership is 100%

Defuzzification: Constructing the Output

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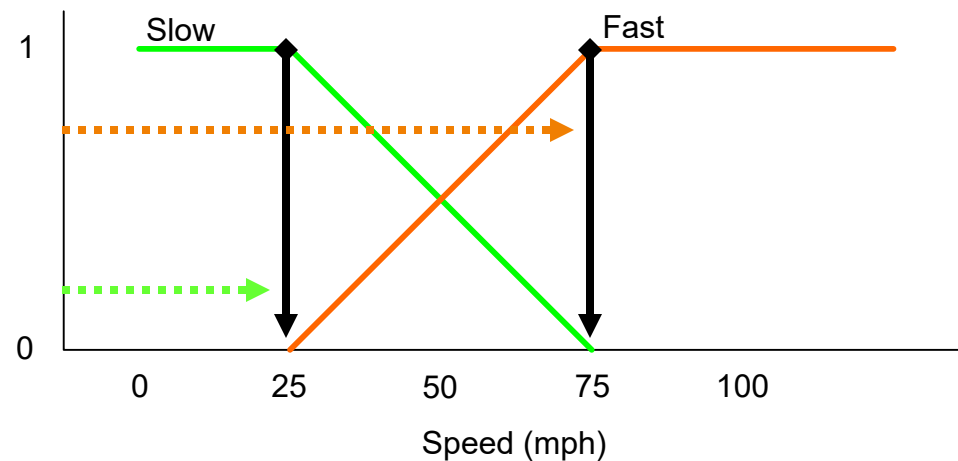


- Speed = weighted mean
= $(2 * 25 + \dots)$

Defuzzification: Constructing the Output

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- Speed is 20% Slow and 70% Fast



- Speed = weighted mean
= $(2*25 + 7*75)/(9)$
= 63.8 mph

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- Fuzzy Logic Control allows for the smooth interpolation between variable centroids with relatively few rules
- This does not work with crisp (traditional Boolean) logic
- Provides a natural way to model some types of human expertise in a computer program

Notes: Drawbacks to Fuzzy logic

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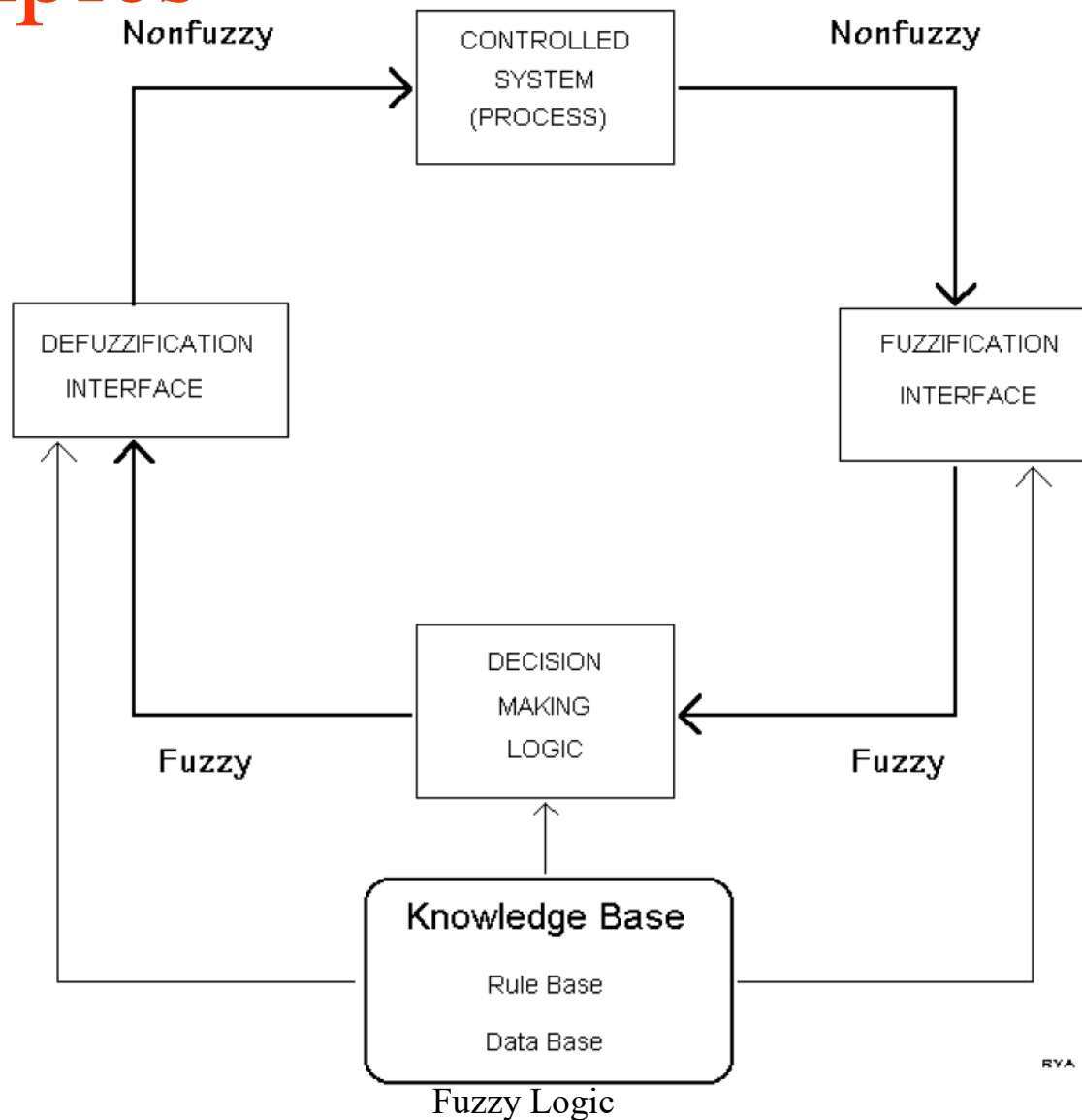
- Requires tuning of membership functions
- Fuzzy Logic control may not scale well to large or complex problems
- Deals with imprecision, and vagueness, but not uncertainty

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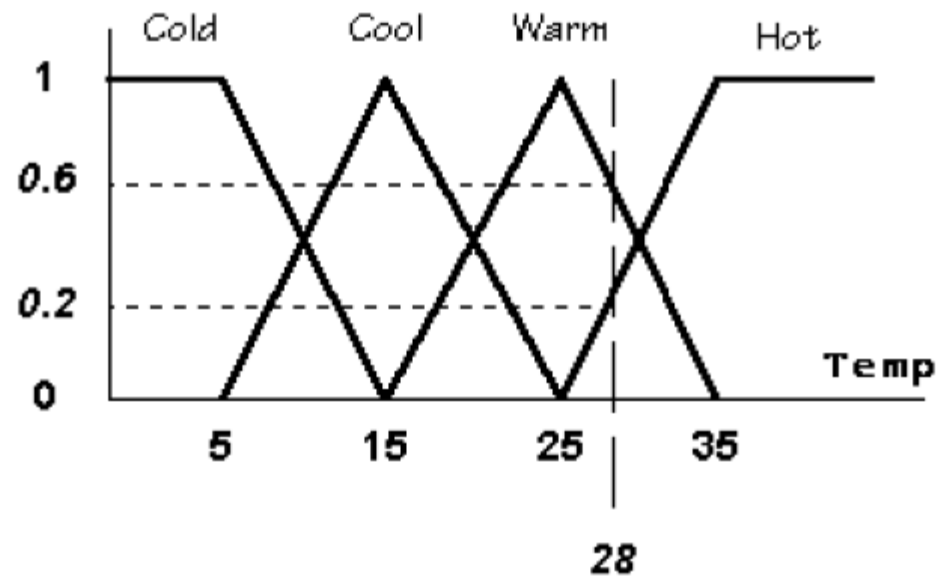
- Fuzzy Logic provides way to calculate with imprecision and vagueness
- Fuzzy Logic can be used to represent some kinds of human expertise
- Fuzzy Membership Sets
- Fuzzy Linguistic Variables
- Fuzzy AND and OR
- Fuzzy Control

Examples



RVA

Fuzzification



Input is the temperature **28**, output is "**0.6 Warm** and **0.2 Hot**"

Rule Evaluation

- 1 : IF temperature IS *cold* THEN fan_speed IS *zero*
- 2 : IF temperature IS *cool* THEN fan_speed IS *low*
- 3 : IF temperature IS *warm* THEN fan_speed IS *medium*
- 4 : IF temperature IS *hot* THEN fan_speed IS *high*

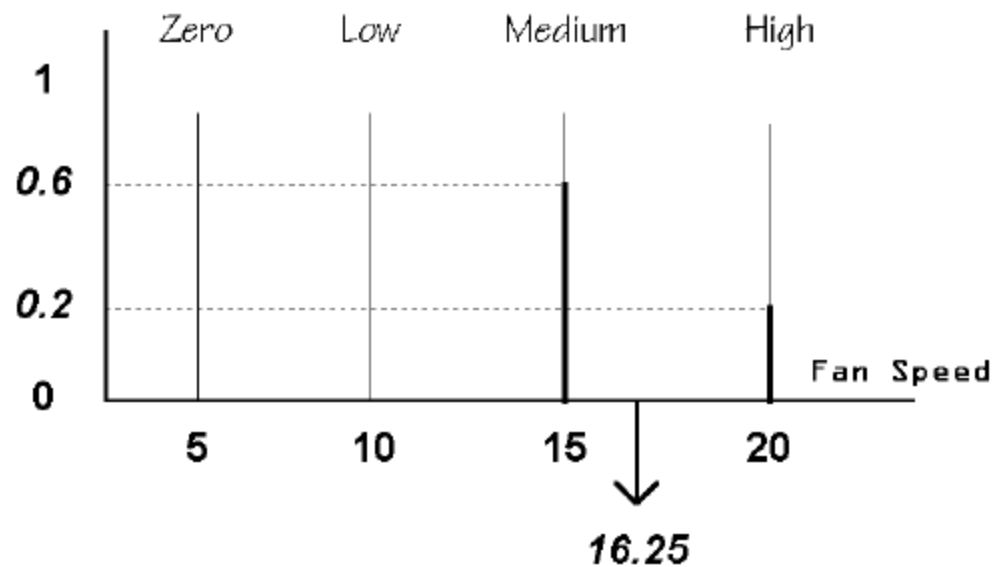
Rule 3 : fan_speed is *medium* with truth value 0.6

Rule 4 : fan_speed is *high* with truth value 0.2

RVA

From the previous **linguistic values** of temperature, **truth values** are computed.

Defuzzification



Finally linguistic values of the control action (fan speed) are translated in numerical value.

Fuzzy Control For a Mobile Robot

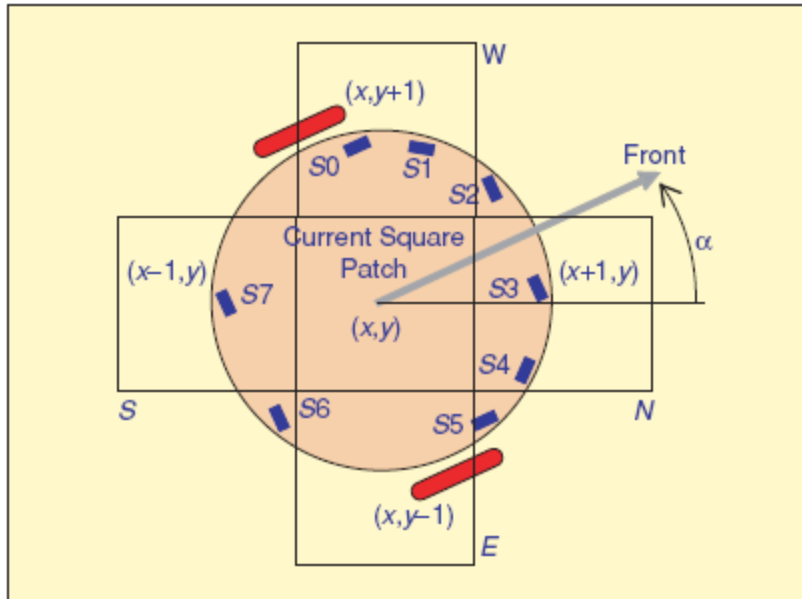


Figure 4. Position and orientation of the infrared sensors and correspondence between arena square patches and exploration matrix.

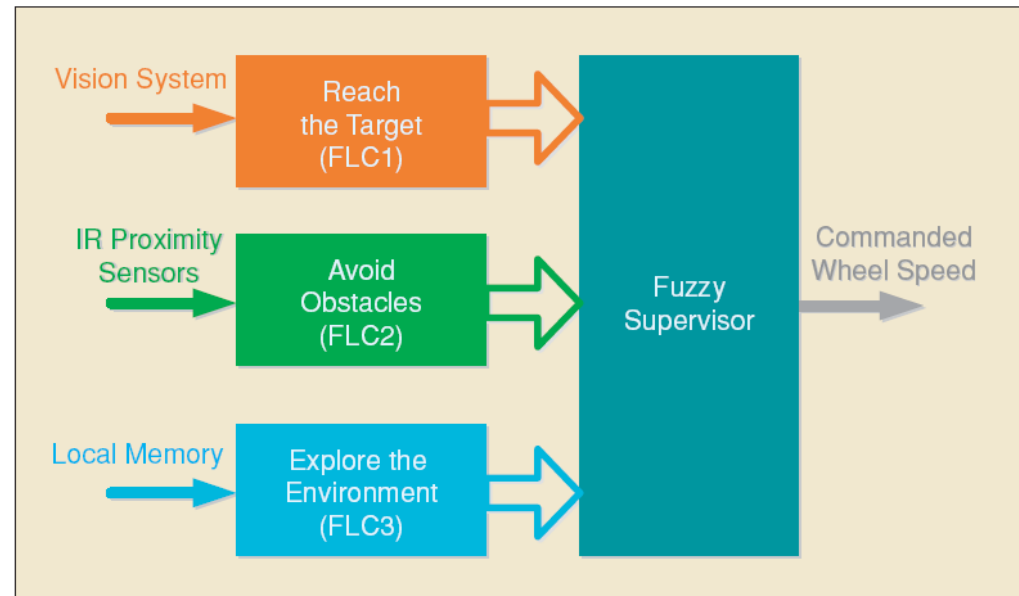


Figure 2. Behavior-based control scheme.

ership



ership

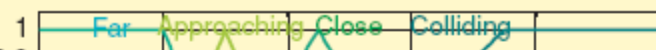


Table 5. Fuzzy supervisor rule table.

INPUTS (logic AND)		OUTPUTS
Max_prox	Exp_Ind	Behavior
Close	Any	Avoid obstacles
Far	Seldom	Reach the target
Far	Often	Explore the environment

(c)

(d)

Figure 3. Input MFs for FLC1 (left-hand), FLC2 (right-hand, upper), and FLC3 (right-hand lower).

Table 1. FLC1 rule

INPUTS (logic AND)		
Dist	Dir	Right speed
Zero	Any	Zero
Near	Center	Positive slow
Far	Center	Positive fast
Any	Left	Negative
Any	Center-left	Negative
Any	Right	Positive fast
Any	Center-right	Positive slow

Table 4. FLC3 rule table.

INPUTS (logic AND)				OUTPUTS	
Delta north	Delta south	Delta east	Delta west	R speed	L speed
Less exp	Less exp	Less exp	Less exp	Positive fast	Positive fast
More exp	Any	Any	Any	Positive fast	Negative fast
Any	More exp	Any	Any	Positive fast	Positive fast
Any	Any	More exp	Any	Positive fast	Positive slow
Any	Any	Any	More exp	Positive slow	Positive fast

FLC2 rule table.

OUTPUTS		
S5	Right speed	Left speed
Any	Negative fast	Aero
Coll	Zero	Negative fast
Any	Negative fast	Positive slow
Close	Positive slow	Negative fast
Far	Positive fast	Positive fast